Assignment: Diabetes Prediction using Machine Learning

In this assignment, you will build a **Machine Learning classification model** using the provided **Diabetes Dataset** to predict whether a patient is likely to have diabetes based on various medical attributes.

Your goal is to implement an end-to-end **Machine Learning classification pipeline**, including data exploration, preprocessing, feature engineering, model building, training, evaluation, and performance analysis using **Python and Scikit-learn**.

This assignment will help you understand:

- End-to-end Machine Learning workflow
- Exploratory Data Analysis (EDA)
- Data preprocessing and handling missing values
- Feature selection and feature engineering
- Splitting data into training and testing sets
- Building and training classification models
- Understanding classification algorithms such as Logistic Regression
- Model evaluation using metrics such as Accuracy
- Interpreting Machine Learning model results and drawing business/analytical insights

Dataset Description

The dataset contains patient health-related attributes that can be used to predict the likelihood of diabetes.

Some important columns include:

Column	Description
Pregnancies	Number of times pregnant
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	Serum insulin level
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes pedigree score
Age	Age of the patient
Outcome	Target Variable (0 = No Diabetes, 1 = Diabetes)

Assignment Tasks

You must complete the following stages as part of this Machine Learning assignment.

Part 1 — Data Preparation

Prepare and preprocess the dataset for Machine Learning model training.

Tasks

1. Load the dataset using Pandas.
2. Explore the dataset:
    a. Shape of the dataset
    b. Column names
    c. Data types
    d. Missing values, if any
    e. Basic statistical summary
3. Separate the dataset into:
    a. Features (X)
    b. Target Variable (y)
4. Perform a Train-Test Split.
5. Perform appropriate preprocessing, if required:
    a. Handle missing values
    b. Encode categorical variables, if any
    c. Scale/standardize numerical features where appropriate

Deliverables

- Dataset exploration summary
- Description of features and target variable
- Preprocessing steps performed
- Training and testing dataset sizes

Part 2 — Machine Learning Model Building

Build a suitable classification model using Scikit-learn.

Tasks

1. Import the required Python and Scikit-learn libraries.
2. Select and implement an appropriate classification algorithm, such as Logistic Regression
3. Train the selected Machine Learning model using the training dataset.
4. Explain why the selected algorithm is suitable for this problem

Deliverables

- Machine Learning model(s) implemented
- Explanation of the selected algorithm

Part 3 — Model Training

Train the Machine Learning model using the prepared training dataset.

Tasks

1. Fit the selected model using the training dataset.
2. Generate predictions on the training dataset.
3. Evaluate the model's performance – accuracy

Deliverables

- Model training output
- Training accuracy
- Compare the training and testing performance of the model.
- Test accuracy
- Precision, Recall, and F1-Score
- Confusion Matrix
- Comparison of training vs. testing performance
- Brief interpretation of the model results

Part 4 — Model Performance Analysis

Analyse the overall performance of the Machine Learning model and draw meaningful conclusions.